

2019 JC2 H2 Physics Preliminary Examination Paper 2 Suggested Solutions

1	(a)	(i)	Applying $v = u + at$ for vertical component, $(-5.5 \sin(33^\circ)) = (5.5 \sin(33^\circ)) + (-9.81)t$ $t = 0.6107 = 0.61 \text{ s}$	1 1
		(ii)	$d = (5.5 \cos(33^\circ)) \times 0.6107 = 2.817 = 2.8 \text{ m}$	1
		(iii)	$\frac{\Delta d}{d} = \frac{\Delta v}{v} + \frac{\Delta t}{t}$ $\frac{\Delta d}{d} = \frac{0.2}{5.5} + 0.02 = 0.056$ Therefore, percentage uncertainty = 5.6%	1 1
		(iv)	$\Delta d = 5.6\% \times 2.817 = 0.2 \text{ m (to 1 s.f.)}$ $d = (2.8 \pm 0.2) \text{ m}$	1
		(v)	$\frac{\text{kinetic energy of the car at its maximum height}}{\text{kinetic energy of the car as it leaves ramp P}}$ $= \frac{\frac{1}{2}m((5.5 \cos(33^\circ))^2)}{\frac{1}{2}m(5.5)^2}$ $= 0.703$	1 1
		(b)	- Straight line from positive value of v_y at $t = 0$ to negative value of v_y - Straight line ends at $t = T$ and final magnitude of v_y greater than initial magnitude of v_y	1 1
2	(a)		Total momentum is constant	1